Progress Report #6: Clean Pipeline Rebuild with Authoritative HSUS Data

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Abstract

This report documents the complete data construction pipeline and estimation results for the immigration–house prices project. We study whether European immigration to US cities during 1910–1930 affected house prices, using a shift-share instrumental variable strategy. The final analysis dataset is an annual panel of 22 metropolitan areas over 21 years (461 city-year observations), constructed from four independent data sources. The 2SLS estimates with city and year fixed effects show a significant negative effect: a one-percentage-point increase in the European immigrant share reduces real house prices by approximately 1.4 log points ($p < 0.05$), with a strong first stage (Kleibergen-Paap $F = 42.7$). However, these effects are not robust to the inclusion of baseline city characteristics interacted with year trends.

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1 Research Question and Identification Strategy

We estimate the causal effect of European immigration on US house prices during the Age of Mass Migration (1910–1930). The key empirical challenge is that immigrants' location choices are endogenous: immigrants may sort into cities with low or declining housing costs, biasing OLS estimates. We address this using a shift-share (Bartik) instrumental variable, following Tabellini (2020).

The instrument predicts city-level immigration using the interaction of two sources of variation: (i) the historical settlement patterns of European immigrant groups across US cities in 1900 (the "shares"), and (ii) annual national immigration flows by country of origin (the "shifts"). The identifying assumption is that the 1900 settlement patterns of immigrant groups affect future house prices *only* through their effect on subsequent immigration flows to those cities.

2 Raw Data Sources

The analysis draws on four independent data sources:

(1) **Historical Statistics of the United States (HSUS), Millennial Edition.** Published by Cambridge University Press (2006), edited by Carter et al. We use Series Ad106–120: "Immigrants by country of last residence – Europe, 1820–1997." The data is downloaded in Excel format from the HSUS online platform. It provides *annual* counts of European immigrants admitted to the United States, disaggregated into 15 country/region columns: Great Britain (Ad107), Ireland (Ad108), Scandinavia (Ad109), Other Northwestern Europe (Ad110), Germany (Ad111), Poland (Ad112), Other Central Europe (Ad113), Former Soviet Union and Baltic States (Ad114), Other Eastern Europe (Ad115), Greece (Ad116), Italy (Ad117), Portugal (Ad118), Spain (Ad119), and Other Europe (Ad120). We use fiscal years 1900–1930. Note: the fiscal year ending date changed over time (June 30 for most of our period).

(2) **IPUMS USA Complete-Count Census Microdata.** We use three extracts:

- *1900 Census* (∼76 million person records): Used to construct the settlement shares—the cross-sectional distribution of each European immigrant group across US cities in 1900.
- *1910, 1920, 1930 Censuses* (∼320 million person records): Used to construct the endogenous variable—annual immigrant arrivals by city, derived from the YRIMMIG (year of immigration) variable reported by foreign-born respondents.

Key variables: BPL (birthplace), METAREA (metropolitan area), NATIVITY (native/foreign-born status), YRIMMIG (year of immigration).

(3) **Lyons et al. (2024) Historical House Price Data.** Individual-level real estate transaction records for US cities, 1890–1935, digitized from historical newspapers and property records. Contains sale prices and rents with property characteristics (rooms, bathrooms, storeys, property type, neighborhood). We use this to construct five house price/rent outcome measures at the city-year level.

(4) **Tabellini (2020) Replication Package.** City-level panel data for US metropolitan areas, 1910–1930 (decennial). We extract: city population in 1910 (used as the denominator for normalizing the instrument and immigrant share), baseline city characteristics in 1900 (log population, log immigration stock) used as controls, and the Tabellini shift-share instrument (used for validation only). The replication data covers 180 observations (60 metro areas $\times$ 3 decades).

3 Data Construction

This section documents, step by step, how the raw data sources are transformed into the final analysis dataset. Each subsection corresponds to a distinct processing step. The code files referenced are in the `codes/build/` directory.

3.1 Step 1: Process HSUS Immigration Flows

Input: HSUS Excel file `Ad106-120.xls` (15 country columns $\times$ 180 years, 1820–1997).

Processing:

1. Read the Excel file. Drop one non-numeric row ("(TQ)" – a transitional quarter entry from when the fiscal year end date changed).

2. Rename the 15 HSUS columns to short variable names (e.g., `NorthwesternEurope_GreatBritain_Ad1` $\rightarrow$ `great_britain`).

3. Filter to fiscal years 1900–1930 (31 rows).

4. Handle missing values: Poland (Ad112) has 20 missing values for years 1900–1919 because Poland was not reported as a separate country before its independence in 1918. These are filled with zero, which is correct because pre-1920 Polish immigration is already captured in the "Other Central Europe" (Ad113) column.

5. **Merge columns into 10 country groups** to match the settlement shares (see Step 2):

   - `other_central` = Poland (Ad112) + Other Central Europe (Ad113). This ensures consistency with the settlement shares, where Polish immigrants (IPUMS birthplace code 455) are classified under "other central" because Poland did not exist as an independent state in the 1900 Census.
   - `other_southern` = Greece (Ad116) + Portugal (Ad118) + Spain (Ad119).
   - The remaining 7 columns are kept as-is: `great_britain`, `ireland`, `scandinavia`, `other_northwestern`, `germany`, `ussr_baltic`, `other_eastern`, `italy`.

6. Validate: the sum of the 10 country-group columns equals the HSUS total (Ad106) within 1.2%. The small residual is the "Other Europe" column (Ad120), which captures minor countries not attributable to any group and has no counterpart in the settlement shares.

7. Save as Stata `.dta` file (31 rows × 12 columns: year, `europe_total`, and 10 country groups).

Output: `us_immigration_by_european_country_1900_1930.dta` — annual national immigration flows by 10 European country groups, 1900–1930.

Code: `04_process_hsus_immigration.py`

3.2 Step 2: Construct 1900 Settlement Shares from IPUMS

Input: IPUMS 1900 complete-count Census extract (`usa_00107.csv.gz`, ∼76 million person records).

Processing:

1. Read the compressed CSV in chunks of 1 million rows (for memory efficiency).

2. Filter to foreign-born Europeans: keep records with NATIVITY $\in \{4, 5\}$ (foreign-born) and BPL (birthplace) codes 400–499 (Europe).

3. Map each IPUMS BPL code to one of 10 country groups using a lookup table:

   - `great_britain`: BPL 410 (England), 411 (Scotland), 412 (Wales), 413–414 (UK, ns)

- `ireland`: BPL 420
- `scandinavia`: BPL 400 (Denmark), 404 (Norway), 405 (Sweden), 406 (Iceland)
- `other_northwestern`: BPL 421 (Belgium), 422 (Luxembourg), 425–426 (Netherlands), 430 (France), 440 (Switzerland)
- `germany`: BPL 453
- `other_central`: BPL 450 (Austria), 451 (Hungary), 452 (Czechoslovakia), 455 (Poland), 457 (Yugoslavia)
- `ussr_baltic`: BPL 401 (Finland), 460 (Russia/USSR), 461 (Latvia), 462 (Estonia), 463 (Lithuania), 465 (Other USSR)
- `other_eastern`: BPL 456 (Romania), 458 (Turkey)
- `italy`: BPL 434
- `other_southern`: BPL 431, 436 (Portugal), 432–433 (Spain), 470 (Greece)

Note: Finland (BPL 401) is classified under `ussr_baltic` (not Scandinavia) to match the HSUS grouping, where Finland is included in "Former Soviet Union and Baltic States." Poland (BPL 455) is classified under `other_central` because Poland was not independent in 1900.

4. Drop records with METAREA = 0 (metropolitan area not identifiable).
5. Tabulate: count immigrants by METAREA × country group (n_{ck}).
6. Compute national totals by country group ($n_k = \sum_c n_{ck}$).
7. Compute settlement shares: $\text{share}_{ck} = n_{ck}/n_k$ — the fraction of country-group-k immigrants nationwide who live in city c.
8. Pivot to wide format: one row per METAREA, 10 columns (`share_great_britain`, ..., `share_other_southern`).
9. Save as `.dta` (62 metro areas × 10 share variables).

Output: `settlement_shares_1900.dta` — the cross-sectional distribution of each European immigrant group across 62 US metro areas in 1900.

Code: `08b_process_ipums_settlement_shares.py`

3.3 Step 3: Construct Endogenous Variable from IPUMS

Input: IPUMS 1910–1930 complete-count Census extract (`usa_00108.csv.gz`, ∼320 million person records).

Processing:

1. Read in chunks. Filter to foreign-born Europeans with valid YRIMMIG (year of immigration) and METAREA > 0.

2. Assign each Census to a non-overlapping year range to avoid double-counting:

   - 1910 Census: arrivals with YRIMMIG ≤ 1910 (years 1900–1910)
   - 1920 Census: arrivals with YRIMMIG $\in [1911, 1920]$
   - 1930 Census: arrivals with YRIMMIG $\in [1921, 1930]$

   Each Census is the "nearest subsequent" observation for its year range, minimizing survival bias (emigrants and deceased are excluded, but the nearest Census minimizes the gap).

3. Count immigrants by METAREA $\times$ arrival year: n_{ct} = number of European immigrants in metro area c who reported arriving in year t.

4. Expand to a balanced panel (all METAREA $\times$ year combinations, 1900–1930). Fill missing counts with zero.

5. Compute 10-year rolling sum for each city:

$$\text{RecentArrivals}_{ct} = \sum_{s=t-9}^{t} n_{cs}$$

   This uses `pandas .transform(lambda x: x.rolling(window=10, min_periods=1).sum())` to maintain correct index alignment within groups.

6. Save `n_arrivals` (annual) and `recent_arrivals` (10-year rolling sum) by METAREA $\times$ year.

Important note on measurement: The YRIMMIG variable exhibits "heaping" at round numbers—years ending in 0 have approximately 4.1 million observations vs. ∼2.2 million for other terminal digits. This reflects respondents rounding their arrival year. The IV strategy absorbs classical measurement error in the endogenous variable, but non-classical heaping could affect the first stage if it is correlated with city characteristics.

Output: `annual_immigration_shares.dta` — annual immigrant arrival counts and 10-year rolling sums, by metro area, 1900–1930.

Code: `08c_process_ipums_endogenous.py`

3.4 Step 4: Construct the Shift-Share Instrument

Inputs:

- Settlement shares from Step 2 (`settlement_shares_1900.dta`, 62 metro areas × 10 groups)
- National flows from Step 1 (`us_immigration_by_european_country_ 1900_1930.dta`, 31 years × 10 groups)
- City population from Tabellini (2020) replication data (1910 Census)

Processing:

1. Load the settlement shares (62 metro areas, wide format with `share_`k variables).
2. Expand to a city-year panel: each metro area gets 31 rows (1900–1930).
3. Merge the annual national flows on year (m : 1 merge, 1,922 matched, 0 unmatched).
4. Verify share–flow alignment: confirm that all 10 country groups have both a `share_`k variable and a flow variable k present in the data. Result: 10/10 matched.
5. Compute the unnormalized instrument:

$$Z_{ct}^{\text{raw}} = \sum_{k=1}^{10} \text{share}_{ck,1900} \times F_{kt}$$

   This is a weighted sum of national flows, where the weights are city c's 1900 settlement shares. All 1,922 observations have non-missing Z^{raw}.

6. Merge city population (1910) from Tabellini replication data on METAREA. One metro area (Salt Lake City, METAREA 716) has no match in the Tabellini data, leaving 1,891 matched observations.

7. Normalize:

$$Z_{ct} = \frac{Z_{ct}^{\text{raw}}}{\text{Pop}_{c,1910}}$$

where $\text{Pop}_{c,1910}$ is the city population from the 1910 Census as reported in Tabellini (2020). 31 observations (Salt Lake City $\times$ 31 years) have missing Z_{ct} due to the unmatched population.

Output: `annual_instrument.dta` — the annual shift-share instrument, 1,922 city-year observations (1,891 with non-missing Z_{ct}).

Code: `07_construct_annual_instrument.do`

3.5 Step 5: Construct House Price Outcomes from Lyons Data

Input: Lyons et al. (2024) individual transaction data (`starting_hhp_data.dta`, 653 MB, ∼2.7 million records).

Processing:

1. *Simple price measures* (`11_prepare_lyons_annual.do`): For each city-year, compute the mean and median of ln(sale price) and ln(rent) across all transactions. Deflate by the CPI (1910 base) to obtain real prices.

2. *Hedonic price indices* (`09_construct_lyons_hpi.do`, `10_construct_lyons_rpi.do`): Following the Lyons et al. "All-in-One" (AIO) specification, estimate per-city hedonic regressions:

$$\ln(\text{price}_{ijt}) = \sum_t \beta_t \cdot \mathbf{1}(t) + \mathbf{X}_{ij}'\gamma + \varepsilon_{ijt}$$

where $\mathbf{X}_{ij}$ includes property characteristics (rooms, bathrooms, storeys, property type, neighborhood fixed effects). The year coefficients $\hat{\beta}_t$ form the hedonic price index (HPI) for each city, normalized to 1 in 1920. The same procedure is applied to rents (RPI).

Output:

- `lyons_city_year_simple_prices.dta`: city-year means/medians of log prices and rents
- `lyons_city_year_hpi.dta`: city-year hedonic sale price index
- `lyons_city_year_rpi.dta`: city-year hedonic rent price index

3.6 Step 6: Merge into Final Analysis Dataset

Inputs: All outputs from Steps 1–5 plus Tabellini baseline controls.

Processing (`12_merge_annual_datasets.do`):

1. Start with the annual instrument (1,922 city-year observations, 62 cities $\times$ 31 years).
2. Merge hedonic HPI ($1:1$ on METAREA $\times$ year): 713 matched, 1,209 master only, 0 using only.
3. Merge hedonic RPI ($1:1$): 713 matched.
4. Merge simple prices ($1:1$): 703 matched.
5. Extract Tabellini baseline controls: for each metro area, take the 1900 values of log city population (`l_citypop900`) and log immigration stock (`l_imm900`) from the Tabellini city dataset. These are time-invariant city characteristics that capture pre-existing city size and immigrant concentration. Merge ($m:1$ on METAREA): 1,891 matched, 31 unmatched (Salt Lake City).
6. Create year-interacted controls: for each year $t \in [1900, 1930]$, generate `l_citypop900_y`t = `l_citypop900` $\times$ $\mathbf{1}(\text{year} = t)$ and similarly for `l_imm900`. These allow baseline characteristics to have year-specific effects on outcomes.
7. Set panel structure: `xtset metarea year`.
8. Restrict to 1910–1930 for the analysis.

Merging the endogenous variable: At the analysis stage, the endogenous variable from Step 3 is merged into the analysis dataset ($1:1$ on METAREA $\times$ year). The immigrant share is then computed as:

$$\text{ImmShare}_{ct} = \frac{\text{RecentArrivals}_{ct}}{\text{Pop}_{c,1910}}$$

Output: `annual_analysis_dataset.dta` — the final analysis-ready panel. After restricting to 1910–1930 and merging the endogenous variable: 1,302 observations (62 cities $\times$ 21 years), of which 461 have complete data on the instrument, immigrant share, and at least one sale price measure.

4 Final Dataset Description

The estimation sample is an annual panel of 22 metropolitan areas over 21 years (1910–1930), yielding 461 city-year observations with complete data on the instrument, endogenous variable, and sale price outcomes (462 for rent outcomes).

Why 22 cities, not 62? The settlement shares cover 62 metro areas from the 1900 Census. However, the Lyons house price data is available for only ∼23 cities, and one of those (Salt Lake City) lacks population data from the Tabellini replication package. The intersection of all four data requirements (settlement shares, national flows, price data, and population) yields 22 cities.

4.1 Summary Statistics

Table 1 reports summary statistics for the estimation sample.

Table 1: Summary Statistics: Estimation Sample (22 Cities, 1910–1930)

Variable	N	Mean	SD	Min	Max
Panel A: Outcome variables (log, real)					
Simple mean log sale price	461	8.157	0.363	7.376	9.191
Simple median log sale price	461	8.149	0.351	7.293	9.149
Log hedonic HPI	461	−0.547	0.228	−1.244	−0.019
Log hedonic RPI	461	−0.797	0.250	−1.543	−0.131
Mean log rent (real)	461	3.198	0.340	2.211	4.431
Number of sale transactions	461	377	266	27	1,418
Panel B: Endogenous variable					
Immigrant share (ImmShare)	461	0.076	0.077	0.001	0.345
Panel C: Instrument					
Shift-share instrument (Z)	461	0.012	0.017	0.0001	0.091

The estimation sample includes 22 metropolitan areas with non-missing values for the instrument, immigrant share, and at least one price measure. All price variables are deflated by CPI with 1910 base year. ImmShare is the 10-year rolling count of European immigrant arrivals (from IPUMS YRIMMIG) divided by 1910 city population. The shift-share instrument Z_{ct} is constructed from 1900 settlement shares × annual HSUS immigration flows, normalized by 1910 city population.

The mean immigrant share is 7.6%, with substantial variation (SD = 7.7 pp), ranging from near-zero to 34.5%. The instrument Z has a mean of 1.2% with a right-skewed distri-

bution (max = 9.1%), reflecting that a few cities received disproportionately large predicted immigration inflows during peak years (1900–1914).

5 Econometric Specification

5.1 OLS

The baseline OLS regression is:

$$y_{ct} = \beta \cdot \text{ImmShare}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \tag{1}$$

where y_{ct} is a log real house price or rent measure for city c in year t, ImmShare_{ct} is the European immigrant share, α_c are city fixed effects (absorbing time-invariant city characteristics), and δ_t are year fixed effects (absorbing common shocks like national business cycles, CPI changes, or federal immigration policy). Standard errors are clustered at the city level.

The coefficient β captures the within-city correlation between immigration and house prices, controlling for common time trends. However, β is biased if immigrants' location choices are correlated with unobserved city-level shocks to housing demand or supply.

5.2 Two-Stage Least Squares (2SLS)

To address the endogeneity of immigrant location choices, we estimate a 2SLS model.

First stage:

$$\text{ImmShare}_{ct} = \gamma \cdot Z_{ct} + \alpha_c + \delta_t + \eta_{ct} \tag{2}$$

where Z_{ct} is the shift-share instrument defined in Section 3.4. The coefficient γ captures how strongly the predicted immigration (based on 1900 settlement patterns $\times$ national flows) predicts actual immigrant shares.

Second stage:

$$y_{ct} = \beta^{IV} \cdot \widehat{\text{ImmShare}}_{ct} + \alpha_c + \delta_t + u_{ct} \tag{3}$$

where $\widehat{\text{ImmShare}}_{ct}$ are the fitted values from the first stage. The coefficient β^{IV} is the local average treatment effect (LATE) of immigration on house prices, identified from variation in immigration driven by historical settlement patterns.

Standard errors: All standard errors are clustered at the metropolitan area level to account for serial correlation within cities and heteroskedasticity. With 22 clusters, this is below the conventional threshold of 30 for reliable asymptotic cluster-robust inference. We note this as a limitation.

Implementation: We use `ivreg2` in Stata with the `partial()` option, which applies the Frisch-Waugh-Lovell theorem to absorb city and year fixed effects before estimation. This is numerically equivalent to including explicit fixed effect dummies but computationally more efficient. The Kleibergen-Paap (KP) Wald rk F-statistic is reported as the weak-instrument diagnostic.

Controlled specification: We also estimate a version replacing year fixed effects with baseline city characteristics interacted with year dummies:

$$y_{ct} = \beta^{IV} \cdot \widehat{\text{ImmShare}}_{ct} + \alpha_c + \mathbf{X}'_{c,1900}\boldsymbol{\lambda}_t + u_{ct} \tag{4}$$

where $\mathbf{X}_{c,1900} = (\ln \text{Pop}_{c,1900}, \ln \text{ImmStock}_{c,1900})'$ and $\boldsymbol{\lambda}_t$ are year-specific coefficients. This allows cities with different initial sizes and immigrant concentrations to follow different price trajectories, absorbing potential confounders correlated with the instrument.

5.3 Reduced Form

The reduced-form regression tests whether the instrument directly predicts house prices:

$$y_{ct} = \pi \cdot Z_{ct} + \alpha_c + \delta_t + \nu_{ct} \tag{5}$$

The reduced-form coefficient π equals the product of the first-stage and second-stage coefficients ($\pi = \gamma \times \beta^{IV}$). Unlike 2SLS, the reduced form does not suffer from weak-instrument bias, making it a useful robustness check.

6 Results

6.1 OLS Estimates

Table 2 reports OLS estimates of equation (1).

All OLS coefficients are negative but statistically insignificant. The point estimates are small (-0.23 to -0.48), suggesting little within-city association between immigration

Table 2: OLS: Immigrant Share on House Prices (City + Year FE)

	Simple Mean	Simple Median	Hedonic HPI	Hedonic RPI	Mean Rent
ImmShare_{ct}	−0.271	−0.327	−0.481	−0.233	−0.331
	(0.501)	(0.525)	(0.530)	(0.488)	(0.646)
N	461	461	461	462	462
R^2	0.889	0.869	0.758	0.741	0.789

Standard errors clustered at metro area level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. All dependent variables in logs, deflated by CPI (1910 base). City and year fixed effects included. 22 clusters.

and house prices after absorbing city and year fixed effects. As discussed below, the 2SLS estimates are substantially larger in magnitude, consistent with OLS being biased toward zero by immigrant sorting into cheaper cities.

6.2 First Stage

Table 3 reports the first-stage regression of ImmShare on the shift-share instrument.

Table 3: First Stage: Immigrant Share on Shift-Share Instrument

	(1) City + Year FE	(2) City FE + Controls
Z_{ct}	1.946^{***}	1.966^{***}
	(0.246)	(0.285)
KP F-statistic	42.70	10.42
Clusters	22	22
N	461	461
R^2	0.850	0.860

Standard errors clustered at metro area level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Dependent variable: ImmShare (10-year European arrivals / 1910 city pop). Column (1) includes city and year FE. Column (2) includes city FE and baseline controls (log 1900 city population, log 1900 immigrant stock) interacted with year dummies (no separate year FE). KP F is the Kleibergen-Paap Wald rk F-statistic from the corresponding 2SLS estimation on the sale price sample.

The first-stage coefficient is approximately 1.95 in both specifications: a one-unit increase

in predicted immigration (as a share of city population) translates into a 1.95 percentage-point increase in the actual immigrant share. The instrument is strong in Column (1), with a KP F-statistic of 42.7, well above the Stock and Yogo (2005) critical value of 16.38 for 10% maximal IV size.

In Column (2), adding baseline controls reduces the KP F to 10.4 — just above the conventional threshold of 10, but substantially weaker. This suggests that the baseline controls (particularly log 1900 immigrant stock × year) absorb a large share of the instrument's identifying variation, since both the controls and the instrument exploit historical immigrant settlement patterns.

6.3 2SLS Estimates

Table 4 reports the main 2SLS results.

Table 4: 2SLS: Effect of Immigration on House Prices

	Simple Mean	Simple Median	Hedonic HPI	Hedonic RPI	Mean Rent
Panel A: City + Year FE					
ImmShare_{ct}	-1.394^{**}	-1.416^{**}	-1.365^{**}	-0.934^{*}	-1.164^{*}
	(0.597)	(0.577)	(0.532)	(0.559)	(0.626)
KP F	42.70	42.70	42.70	42.97	42.97
N	461	461	461	462	462
Panel B: City FE + Baseline Controls × Year					
ImmShare_{ct}	0.057	0.104	0.943	0.397	-0.463
	(0.809)	(0.761)	(0.760)	(0.878)	(1.041)
KP F	10.42	10.42	10.42	10.47	10.47
N	461	461	461	462	462

Standard errors clustered at metro area level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. All dependent variables in logs, deflated by CPI (1910 base). ImmShare instrumented by Z_{ct}. 22 clusters in all specifications. Panel A: city and year FE. Panel B: city FE and baseline controls (log 1900 city pop, log 1900 immigrant stock) × year dummies.

Panel A (city + year FE): All five 2SLS coefficients are negative and statistically significant at the 5% level (sale prices) or 10% level (rents). The estimates are consistent across price measures: approximately −1.4 for sale prices (simple mean, median, hedonic HPI) and

-0.9 to -1.2 for rents. A one-percentage-point increase in the European immigrant share reduces real sale prices by approximately 1.4 log points.

Panel B (with controls): When baseline controls are added, all effects become statistically insignificant and change sign for most outcomes. The standard errors also increase substantially. The KP F-statistic drops to $\sim$10.4, indicating a much weaker first stage.

Interpretation: The instability across panels is the central interpretive challenge. Two readings are possible: (i) the controls absorb omitted confounders that drove the Panel A results (implying no causal effect), or (ii) the controls are collinear with the instrument and absorb its identifying variation (implying Panel B is underpowered, not contradictory). The substantial drop in the F-statistic supports the second reading, but we cannot definitively distinguish the two.

6.4 Reduced Form

Table 5 reports the reduced-form estimates.

Table 5: Reduced Form: Effect of Instrument on House Prices

	Simple Mean	Simple Median	Hedonic HPI	Hedonic RPI	Mean Rent
Panel A: City + Year FE					
Z_{ct}	-2.532^{***}	-2.572^{***}	-2.479^{**}	-1.695	-2.113^{*}
	(0.913)	(0.895)	(0.964)	(1.071)	(1.252)
N	461	461	461	462	462
Panel B: City FE + Baseline Controls × Year					
Z_{ct}	0.092	0.167	1.512	0.637	-0.742
	(1.404)	(1.330)	(1.250)	(1.474)	(1.833)
N	461	461	461	462	462

Standard errors clustered at metro area level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. All dependent variables in logs, deflated by CPI (1910 base). 22 clusters.

The reduced-form results mirror the 2SLS pattern. Panel A shows significant negative effects: cities that received larger predicted immigration inflows experienced lower house prices. The reduced form is useful because it does not suffer from weak-instrument bias.

Internal consistency check: The Wald ratio (reduced form / first stage) provides an independent estimate of β^{IV}: $-2.532/1.946 = -1.301$, which is close to the 2SLS estimate of -1.394. The small discrepancy reflects sample differences (the first stage reported in Table 3 uses a slightly different sample than the one embedded in the 2SLS estimation).

7 Discussion and Limitations

1. **Sensitivity to controls.** The most important caveat: 2SLS effects are significant in the FE-only specification but vanish when baseline controls are added. Distinguishing whether the controls absorb confounders or identifying variation requires further investigation—for example, adding controls one at a time, or using alternative instruments.

2. **Few clusters.** With only 22 metropolitan areas, cluster-robust standard errors may understate true uncertainty. Wild cluster bootstrap (e.g., Cameron et al., 2008) should be implemented to verify statistical significance.

3. **Magnitude.** The 2SLS coefficient of -1.4 implies that a one-percentage-point increase in immigrant share reduces prices by 1.4%. At the mean immigrant share of 7.6%, the implied cumulative effect is -10.6%. While large, this is not implausible for the historical context, when anti-immigrant sentiment was intense and restrictive quotas were enacted (1921, 1924).

4. **YRIMMIG measurement error.** The endogenous variable relies on self-reported year of immigration, which exhibits heaping at round numbers.

5. **Census boundary discontinuities.** The endogenous variable splices data from three Censuses. Differential survival/emigration rates could create artificial breaks at Census boundaries.

6. **Exclusion restriction.** The instrument is valid only if 1900 settlement patterns affect house prices exclusively through subsequent immigration. This could be violated if ethnic enclaves have persistent direct effects on housing markets.

7. **External validity.** The sample covers 22 large metropolitan areas with Lyons price data. Results may not generalize to smaller cities or non-urban areas.

8 Human–AI Collaboration

Human researcher (Guohui Jiang):

- Located and downloaded the HSUS data from Cambridge University Press (replacing previous digitization attempts)
- Directed the full pipeline audit before re-running analysis
- Required step-by-step validation with independent review agents
- Determined the overall research strategy and interpretation

Claude Code (AI assistant):

- Conducted systematic audit of the entire build and analysis pipeline, identifying 4 bugs
- Wrote the HSUS processing script (`04_process_hsus_immigration.py`)
- Fixed: Poland flows excluded from instrument, BPL dictionary collision, IV estimator compatibility, controls collinearity
- Ran the full pipeline with step-by-step validation agents at each stage
- Drafted this progress report and research log entries

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